

AQUABLEND

Deakin University — Capstone Project

On the Search for Cluster Analysis

A survey and evaluation of clustering techniques for AquaBlend data

Prepared by

Name	ID	Role
Bao Minh Tran	s224236373	Bridge Lead, Optimisation & Analysis / AI

Reviewed by

Name	ID	Role
<i>Full Name</i>	<i>sXXXXXXXXXX</i>	<i>Reviewer / Supervisor</i>

Version Draft v0.1
Status Draft
Date July 30, 2026

Revision history

Version	Date	Author	Summary of changes
v0.1	July 30, 2026	B. M. Tran	Initial document structure.

Abstract

One paragraph covering: the problem this document addresses, the clustering techniques surveyed, how they were evaluated, and the headline finding or recommendation for AquaBlend.

[TODO: write once the first technique is complete]

Contents

Revision history	i
Notation and acronyms	vi
1 Introduction	1
1.1 Motivation	1
1.2 Objectives	1
1.3 Scope and limitations	1
1.4 Document structure	2
2 Background	3
2.1 Cluster analysis in brief	3
2.2 Families of clustering techniques	4
2.3 Related work	4
3 Data	6
3.1 Source and provenance	6
3.2 Features	6
3.3 Preprocessing	6
3.4 Exploratory summary	6
4 Evaluation protocol	7
4.1 Experimental setup	7
4.2 Validation metrics	7
4.3 Choosing the number of clusters	7
4.4 Interpreting and naming clusters	7
4.5 Threats to validity	7
5 Exploratory Data Analysis and Data Mining Pipeline	8
5.1 Exploratory Data Analysis (EDA) methods	8
5.2 Data Mining pipeline	9
6 Discussion on Dimensionality problems	10
6.1 Curse of Dimensionality	10
6.2 Manifold hypothesis and Manifold learning	10
6.3 Linear Dimensionality Reduction techniques	10
6.4 Nonlinear Dimensionality Reduction techniques	10

7	Clustering methods	11
7.1	Similarity measures and validation	11
7.2	Hierarchical clustering methods	12
7.2.1	<NAME>	12
7.3	Partition-based clustering methods	13
7.4	Hybrid clustering methods	14
8	Comparative discussion	15
8.1	Side-by-side results	15
8.2	Qualitative comparison	15
8.3	Recommendation	15
9	Conclusion and future work	16
9.1	Conclusion	16
9.2	Future work	16
A	Reproducibility	17
A.1	Environment	17
A.2	Running the analysis	17
B	Supplementary results	17
	Acknowledgements	18
	References	19

List of Figures

List of Tables

1	Validation metrics reported for every technique.	7
2	Validation metrics across all techniques surveyed.	15

Notation and acronyms

Acronyms

Acronym	Expansion
UML	Unsupervised Machine Learning
HCA	Hierarchical Cluster Analysis
DHC	Divisive Hierarchical Clustering
AHC	Agglomerative Hierarchical Clustering
NC	Number of Cluster
DR	Dimensionality Reduction
SSE	Sum of Squared Error

Mathematical notation

Symbol	Meaning
$\mathbf{X}$	Matrix representation of the dataset, rows for samples and columns for features
n	Number of observations
d	Number of features
$\mathcal{C}$	Set of clusters
$\mathcal{C}_i$	i^{th} -cluster of $\mathcal{C}$
$ \mathcal{C} $	Number of clusters
$d(\cdot, \cdot)$	Distance / dissimilarity measure

Conventions

State document-wide conventions here, e.g. vectors are column vectors and set in bold, indices i range over observations and j over clusters, all distances are Euclidean unless stated otherwise, software versions and random seeds used for every run.

1 Introduction

1.1 Motivation

Arising from the desire of improving the project proposal, allowing the core mathematical models to become more proactive, rather than being reactive by nature. In addition, realising the potential of the AquaBlend dataset can be further explored and studied to improve the performance of future Mixed-Integer Nonlinear Programming (MINLP) model, along with insightful patterns undiscovered in the dataset, which is impactful in future research on related problems about water systems based in Melbourne, and possibly adopted to study water systems in other locations.

This document is made to accomodate the needs of summarizing the deliverables of the codebase provided in the project's repository and solid theoretical and practical foundations for exploring and analysing AquaBlend dataset, to facilitate a broad range of functionalities of the project deliverables, including but not limited to, Time-series analysis, Demand Forecasting, Anomaly Detection and Scenarios Generation.

1.2 Objectives

This section outlines questions addressed in this document.

1. What potential clustering methods and dimensionality reduction techniques can be applied to specifically study the AquaBlend dataset?
2. How well the selected methods perform on the AquaBlend dataset subject to certain standard evaluation metrics/scores?
3. Can there be an appropriate Exploratory Data Analysis and Data Mining pipeline to discover patterns in the AquaBlend dataset? If so, can it be standardised to apply to related problems? Otherwise, why is it impossible?
4. Is “Manifold Hypothesis” valid in the dataset? If so, what can be done to improve the clustering results? Otherwise, is there strong statistical evidence attained from the AquaBlend dataset?

1.3 Scope and limitations

Cluster Analysis is a broad research area, thereby we limit ourselves to discussing on specific methods and problems associated, mentioning other methods without going in details. However, this document will be extended to include more clustering methods in the upcoming work.

For the current version, we limit ourselves to those methods and techniques particularly useful to uncover knowledge from AquaBlend dataset. In recent future, we shall improve the pipeline to become adaptable to discover knowledge from relevant datasets aiming to optimise Operational Water Resources Management sector.

More importantly, this document is not a comprehensive survey of all existing studies of Cluster Analysis, and thereby, there may exist more effective and appropriate techniques yet to mentioned within this document. Hence, this document can only be used as a reference for a collection of potential methods, and how they can be practically implemented to solve certain problems.

1.4 Document structure

The document is organised as follows.

1. Survey recent development in Cluster Analysis, see **Sect. 2**.
2. Introduce dataset used (see **Sect. 3**) and experimental setup to benchmark different clustering strategies (see **Sect. 4** and **App. A**).
3. Deliver the Exploratory Data Analysis process and Data Mining pipeline conducted, see **Sect. 5**.
4. Discuss on various problems associated with data dimensionality and possible solutions, see **Sect. 6**.
5. Summarise the theory behind the implementation of clustering methods, see **Sect. 7**.

2 Background

2.1 Cluster analysis in brief

Clustering analysis refers to the process of exploring the dataset to uncover the latent structure and underlying relationships among data points, commonly known as unsupervised machine learning (UML) in the AI community, and essentially important in data mining and knowledge discovery [8]. It comprises of all clustering methods that are capable of handling (usually) high-dimensional data and grouping them by the utilisation of several (usually) quantitative measurements metrics, see [5] for a brief history of development. In this document, we shall refer to a clustering method as [11]:

- Minimising within-cluster (intra-cluster) distance/similarity.
- Maximising intra-cluster distance/similarity.
- Measurement of distance/similarity must be well-defined and practical.

Formally, let us claim the definitions of a valid mathematical metric and a clustering method.

Definition 2.1 (A mathematical metric [7]). Given two points $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, a (mathematical) metric is given by $d : \mathbb{R}^n \times \mathbb{R}^n \mapsto \mathbb{R}$, which satisfies:

- Identity and Positivity: $d(\mathbf{x}, \mathbf{y}) > 0$ if $\mathbf{x} \neq \mathbf{y}$, and $d(\mathbf{x}, \mathbf{y}) = 0 \Leftrightarrow \mathbf{x} = \mathbf{y}$
- Symmetry: $d(\mathbf{x}, \mathbf{y}) = d(\mathbf{y}, \mathbf{x})$
- Triangle inequality: $d(\mathbf{x}, \mathbf{y}) \leq d(\mathbf{x}, \mathbf{z}) + d(\mathbf{z}, \mathbf{y}), \forall \mathbf{z} \in \mathbb{R}^n$

Definition 2.2 (Clustering method [8]). Given a dataset $\mathbf{X} \triangleq \{\mathbf{x}_i \in \mathbb{R}^n\}_{i=1}^m$. A clustering method is defined as $\phi_d : \mathbf{X} \mapsto \mathcal{C}$, such that:

$$\bigcap_{i=1}^{|\mathcal{C}|} \mathcal{C}_i = \emptyset \quad \text{and} \quad \bigcup_{i=1}^{|\mathcal{C}|} \mathcal{C}_i = \mathbf{X} \quad (1)$$

and ϕ_d is (usually) aiming to:

$$\min \sum_{i=1}^{|\mathcal{C}|} \sum_{\substack{\mathbf{x}_j, \mathbf{x}_k \in \mathcal{C}_i \\ \mathbf{x}_j \neq \mathbf{x}_k}} d(\mathbf{x}_j, \mathbf{x}_k) \quad \text{and} \quad \max_{1 \leq i < j \leq |\mathcal{C}|} \left\{ \min_{\substack{\mathbf{x}_i \in \mathcal{C}_i \\ \mathbf{x}_j \in \mathcal{C}_j}} d(\mathbf{x}_i, \mathbf{x}_j) \right\} \quad (2)$$

where $d(\cdot, \cdot)$ is a *dissimilarity measure*, need not to be a metric as in **Def. 2.1**.

Although different literatures proposing different claims of standard process of cluster analysis [5], though sharing a lot in common, we shall adopt the process written in [11].

1. Feature extraction and selection: opt out representative features of the dataset.
2. Clustering algorithm design: select appropriate methods for the data characteristics and problems' constraints.
3. Cluster evaluation/validation: quantify the performance of clustering results by use of validity measures.

4. Result interpretation: explain what drives the method to identify the patterns.

Looking from the above definition and formulation of clustering methods, several issues have been identified, which are (mostly) subject to dataset and nature of clustering methods [8]. Regarding dataset, the Curse of Dimensionality leads to indistinguishable distances between points in high-dimensional space [6], which degrades the capability of dissimilarity measure used in the methods. Other hurdles are, including but not limited to, size of dataset, noise/outliers, and mixed data type. Regarding methods, many of them requires hyperparameter tuning, which is subjective and data-dependent, also but not limited to, computation complexity, convergence speed, and local optima convergence. Therefore, the upcoming clustering techniques have been developed to mitigate some of issues associated with the problems at some extent.

2.2 Families of clustering techniques

This section serves as a brief list of methods' names, for details what each family include, refer to **Sect. 7**. There is no universal agreement on how clustering methods are classified, several articles, e.g. [10] [2] [5] and books, e.g. [9] and [4] rely on different criteria to divide these clustering methods into groups depending on strategies implemented. In this document, we adopt the philosophy in [8] to classify the methods into two major groups.

Hierarchical clustering Hierarchical Cluster Analysis (HCA) offers multi-level partitions of the dataset, usually represented using a dendrogram, and one can select a threshold to obtain different clustering outcomes. It is further categorised into two groups, i.e. a top-down strategy, known as Divisive Hierarchical Clustering (DHC), and a bottom-up strategy, known as Agglomerative Hierarchical Clustering (AHC). Although differing in construction approaches, they share same components, including dissimilarity measure and linkage criterion. However, DHC is more favorable in practices due to its efficiency.

Partition-based clustering Partitional clustering or iterative relocation algorithms focus on diving the dataset into (nearly-)homogenous groups without the needs of hierarchical structure, aiming to optimise certain criteria functions, can be understood as those given in **Def. 2.2**. It is further categorised into two groups, i.e. hard/crisp clustering and soft/fuzzy clustering. In lieu of considering crisp and fuzzy clustering as two primary groups within partitional methods as in [8] [2], we shall classify crisp clustering methods in more thorough manners and keep fuzzy clustering method as is. Hence, the family of partition-based methods, including but not limited to, graph-theoretic, subspace, density-based, model-based, search-based, SSE-optimised, and miscellaneous clustering methods.

Hybrid clustering this family of method comprises those techniques that are mixture of the other techniques, irrespective of what family they reside in.

In reality, clustering can have any arbitrary shape and distribution, this fact underpins distinct classification of the clustering methods. In brief, there are certain types of clusters that have been extensively studied, e.g. well-separated, prototype-based, graph-based, density-based, and shared-property. For detailed visualisations of these clusters in 2D-space, see [9].

2.3 Related work

Cluster analysis is an fundamental phase in knowledge discovery process, and thus, can be applied to a broad range of real-world applications, e.g. bioinformatics [10], health research [1], and

network-related problems [3]. Nonetheless, we limit ourselves to revisit recent work tackling issues and demands associated with water systems in the past few years. **unfinished**.

- List applications that rely on cluster analysis (better to have something related to the project)
- Up to our knowledge, no recent cluster analysis papers on water system

3 Data

3.1 Source and provenance

Where the data comes from, who owns it, which extract or version is used, and the date it was pulled.

3.2 Features

Table of features: name, type, unit, range, and whether the feature is used for clustering or held out for interpretation.

3.3 Preprocessing

Cleaning, missing-value handling, encoding, scaling, and dimensionality reduction — everything applied before clustering, in the order applied, so the pipeline is reproducible.

3.4 Exploratory summary

Distributions, correlations, and anything that constrains the choice of technique (cluster shape, density, outliers, scale).

4 Evaluation protocol

4.1 Experimental setup

Hardware, software versions, random seeds, and the number of restarts per run. Identical for every technique.

4.2 Validation metrics

The internal and external validity measures reported for every technique, with the definition of each given once here.

Table 1: Validation metrics reported for every technique.

Metric	Range	Interpretation
<i>Metric</i>	<i>[a, b]</i>	<i>What a high or low value means</i>

4.3 Choosing the number of clusters

The procedure applied consistently wherever k must be fixed in advance.

4.4 Interpreting and naming clusters

How a resulting partition is turned into something the business can act on, and who signs off on the naming.

4.5 Threats to validity

Known weaknesses of this protocol and how they are mitigated.

5 Exploratory Data Analysis and Data Mining Pipeline

5.1 Exploratory Data Analysis (EDA) methods

5.2 Data Mining pipeline

6 Discussion on Dimensionality problems

6.1 Curse of Dimensionality

6.2 Manifold hypothesis and Manifold learning

6.3 Linear Dimensionality Reduction techniques

6.4 Nonlinear Dimensionality Reduction techniques

7 Clustering methods

Refer to **Sect. 2** for details of cluster analysis, families of methods, and important properties.

7.1 Similarity measures and validation

7.2 Hierarchical clustering methods

7.2.1 <NAME>

Overview *What the algorithm does, in one paragraph, and the family it belongs to (see Section 2.2).*

Assumptions and applicability *What the algorithm assumes about cluster shape, density, scale and noise; the conditions under which it is a sensible choice for the data described in Section 3.*

Formulation *Objective function and update rule, using the symbols declared in the notation table. Cite the original source.*

Hyperparameters and tuning *Each hyperparameter, the range searched, and the value selected — with the reason.*

Complexity *Time and space complexity, and what that means at AquaBlend data volumes.*

Application to AquaBlend data *Implementation, any deviations from the shared preprocessing pipeline, and why they were necessary.*

Results *Metrics from Section 4.2, cluster profiles, and supporting figures.*

Limitations *Where this method failed or would be expected to fail on this data.*

Summary *Two or three sentences: the verdict on this method, carried into the comparison section.*

7.3 Partition-based clustering methods

7.4 Hybrid clustering methods

8 Comparative discussion

8.1 Side-by-side results

Table 2: Validation metrics across all techniques surveyed.

Technique	Metric 1	Metric 2	Runtime
<i>Technique one</i>	—	—	—

8.2 Qualitative comparison

Trade-offs the metrics do not capture: interpretability, stability across seeds, sensitivity to preprocessing, operational cost.

8.3 Recommendation

Which technique AquaBlend should adopt, under what conditions, and what would change the recommendation.

9 Conclusion and future work

9.1 Conclusion

The answer to each objective listed in Section 1.2, in the same order.

9.2 Future work

Techniques not yet covered, open questions, and what the next contributor to this document should pick up first.

A Reproducibility

A.1 Environment

Language, library and package versions, and how to recreate the environment.

A.2 Running the analysis

Commands to reproduce every result in this document, and where the code and outputs live in the repository.

B Supplementary results

Full metric tables, additional figures, and parameter sweeps that would interrupt the main narrative.

Acknowledgements

Thank the people and bodies that contributed: teammates who supplied or cleaned the data, reviewers, supervisors and unit staff, and any external dataset or software provider whose licence requires attribution. Name the contribution, not just the person.

Note here any use of generative AI tools, stating what they were used for and how the output was verified.

References

- [1] Bei Dou, Bert Vandenberk, and Philip Moons. Cluster analysis: An overview of traditional and novel approaches for health researchers. *European Journal of Cardiovascular Nursing*, 2026. Advance online publication, article zvag100.
- [2] Absalom E. Ezugwu, Abiodun M. Ikotun, Olaide O. Oyelade, Laith Abualigah, Jeffery O. Agushaka, Christopher I. Eke, and Andronicus A. Akinyelu. A comprehensive survey of clustering algorithms: State-of-the-art machine learning applications, taxonomy, challenges, and future research prospects. *Engineering Applications of Artificial Intelligence*, 110:104743, 2022.
- [3] Ali Noori Gatea, Haider Sh. Hashim, Hamid Ali Abed AL-Asadi, Didem Kıvanç Türeli, Zaid Ameen Abduljabbar, and Vincent Omollo Nyangaresi. Clustering in data analysis: Comprehensive insights into techniques and challenges. In Radek Silhavy and Petr Silhavy, editors, *Software Engineering: Emerging Trends and Practices in System Development*, pages 108–124, Cham, 2026. Springer Nature Switzerland.
- [4] Jiawei Han, Micheline Kamber, and Jian Pei. *Cluster Analysis: Basic Concepts and Methods*, chapter 10. The Morgan Kaufmann Series in Data Management Systems. Morgan Kaufmann, Waltham, MA, 3 edition, 2011.
- [5] Adam Jaeger and David Banks. Cluster analysis: A modern statistical review. *WIREs Computational Statistics*, 15(3):e1597, 2023.
- [6] Dehua Peng, Zhipeng Gui, and Huayi Wu. Interpreting the curse of dimensionality from distance concentration and manifold effect, 2025.
- [7] Walter Rudin. *Principles of Mathematical Analysis*. International Series in Pure and Applied Mathematics. McGraw-Hill, New York, 3 edition, 1976.
- [8] Jaswinder Singh and Damanpreet Singh. A comprehensive review of clustering techniques in artificial intelligence for knowledge discovery: Taxonomy, challenges, applications and future prospects. *Advanced Engineering Informatics*, 62:102799, 2024.
- [9] Pang-Ning Tan, Michael Steinbach, Anuj Karpatne, and Vipin Kumar. *Cluster Analysis: Basic Concepts and Algorithms*, chapter 7. Pearson Education, New York, NY, 2 edition, 2019.
- [10] Aasim Ayaz Wani. Comprehensive analysis of clustering algorithms: exploring limitations and innovative solutions. *PeerJ Computer Science*, 10:e2286, 08 2024.
- [11] Dongkuan Xu and Ying jie Tian. A comprehensive survey of clustering algorithms. *Annals of Data Science*, 2:165 – 193, 2015.